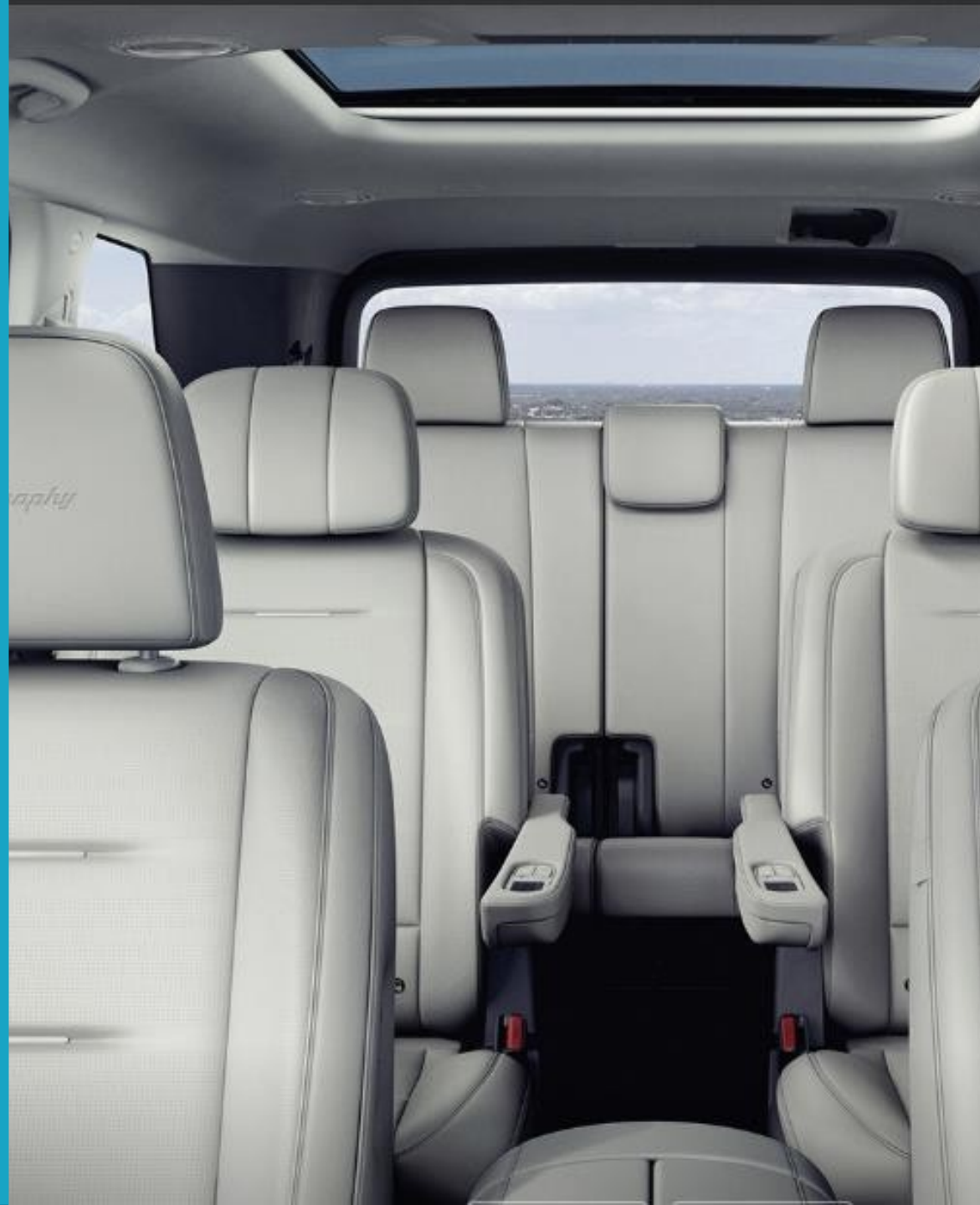


Automotive fastening systems

Corporate and technical profile - Hook and
Twin Lock 1022

TECHNICAL SALES PRESENTATION



ELCRO focuses on automotive fastening



2017

Company founded in
February 2017

Ulsan

Korea

Manufacturing location
listed in the source profile

Core focus

Hook-and-loop fasteners for seat assembly and reclosable attachment systems for automotive interiors.

Development progressed from hook to Twin Lock

2017



Mushroom hook

1440 mm production
method reported

2022



Twin Lock

Manufacturing-method
patent application
reported

2024



Quality system

ISO 9001:2015
certification reported

2025



Twin Lock

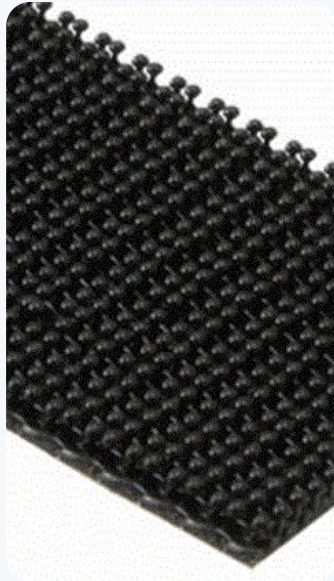
Product development
reported complete

Milestones are company-reported and should be supported with certificates or filing records before external release.

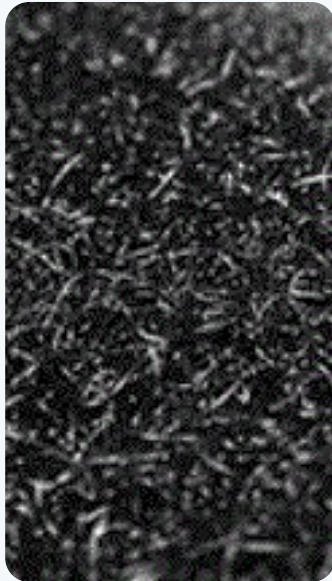
The portfolio spans foam, trim and interiors



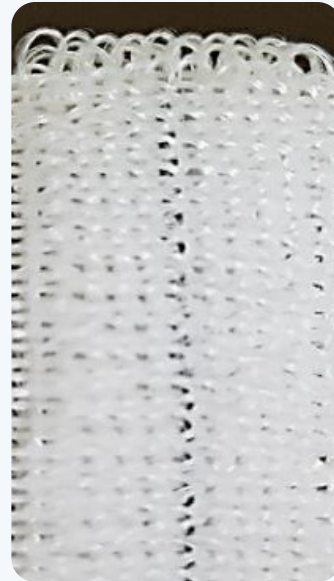
**Mushroom
hook**



Plastic hook



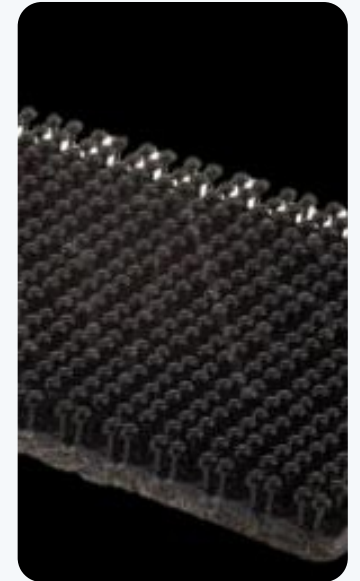
Textile loop



**Automotive
loop**



Cover hook



Twin Lock

The product family supports mold-in retention, trim mating and reclosable interior attachment.

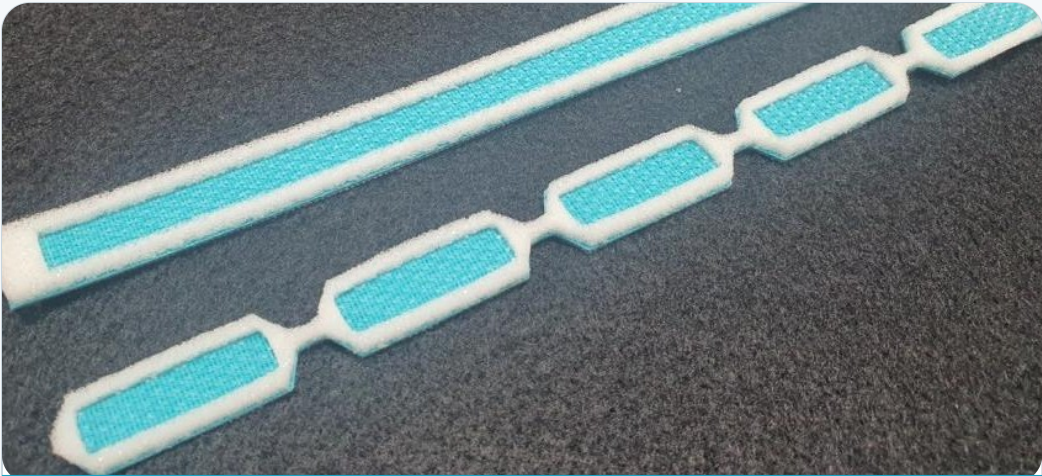
Mold-in hook creates the foam-to-trim interface



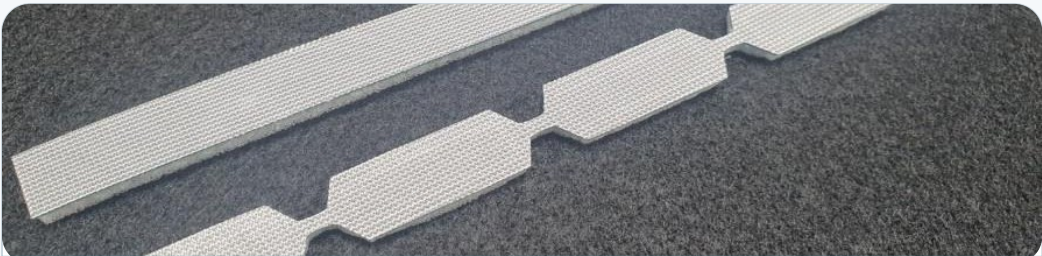
INTERFACE

- Hook is positioned in the foam mold
- Loop is attached to the seat trim
- Engagement locates and retains the cover

Non-cover hook uses five functional layers



Front construction

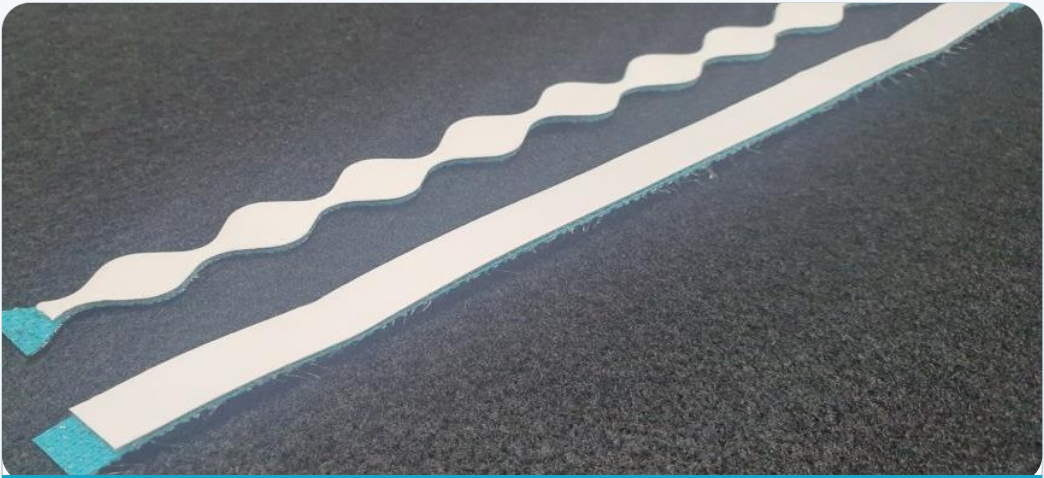


Back construction

Layer / item	Source-reported specification
Hook	Polypropylene mushroom profile
Base	Polyester
Backing	PET film + Fe powder + polyester mesh
Thickness	4.5 +/- 0.1 mm
Dimensional tolerance	+/- 0.1 mm, length and width

Fe powder is listed in the construction. Magnetic mold-positioning performance still requires a documented test.

Cover hook adds protective and mating layers



Cover hook - front



Cover hook - back

Functional group	Source-reported construction
Hook	Polypropylene mushroom + polyester base
Cover	Nylon loop + paper base
Backing	Fe powder + nylon backing
Thickness	4.5 +/- 0.1 mm
Dimensional tolerance	+/- 0.1 mm, length and width

The cover construction is intended to protect the hook during foam molding while preserving the mating interface.

Source tests report 17 kgf and 23 kgf



17

kgf

Hook to loop
10 x 100 mm specimen

23

kgf

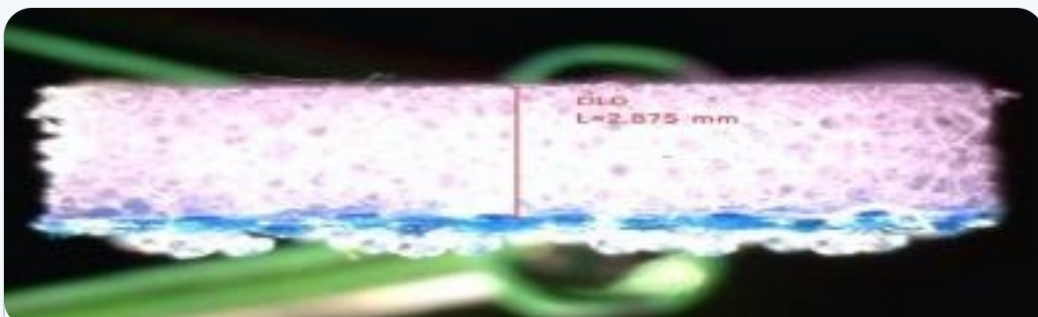
Hook to seat pad
10 x 100 mm specimen

Source specifications

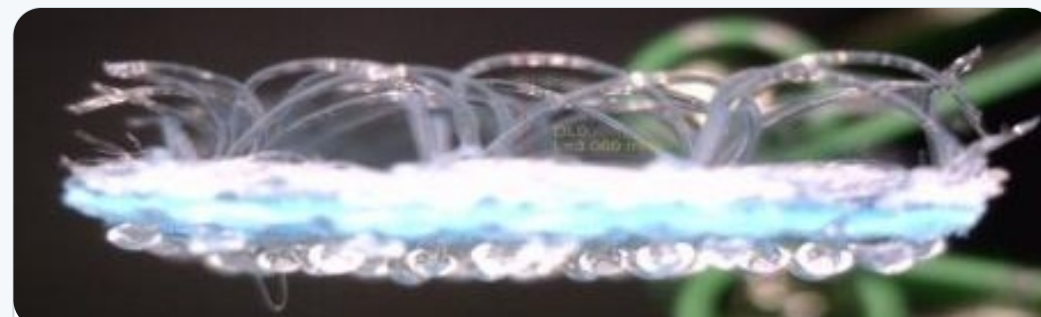
15 +/- 3 kgf hook-to-loop
22 +/- 3 kgf hook-to-pad

Confirm pull direction, rate, conditioning and equipment calibration before customer comparison.

Foam trials show deeper mechanical integration



Cross-section A



Cross-section B



Competitor example



ELCRO example

The measurement basis, sample count and foam-process conditions must be confirmed before publishing a quantified comparison.

Reliability evidence uses Hyundai-Kia MS tests

Test	Standard	Requirement	Source result
Interior flammability	MS 300-08	≤ 80 mm/min	Approved
Interior odor	MS 300-34	Grade 3 or higher	Approved
VOC	MS 300-55	TVOC / HCHO < 2 ppm	Approved
Restricted heavy metals	MS 201-02	Not detected	Approved



These source-reported MS results do not independently establish FMVSS 302 compliance.

Twin Lock enables same-side attachment

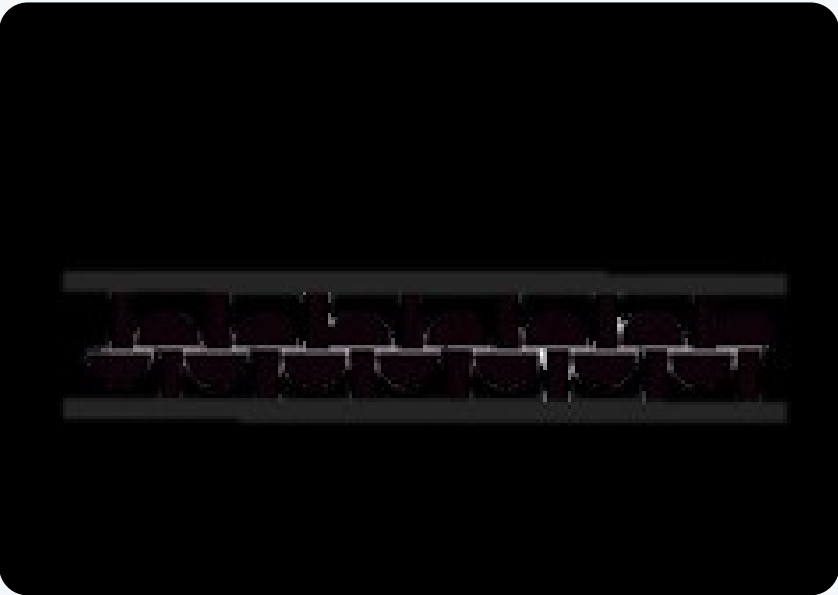


DESIGN INTENT

The source deck describes a fastener that engages without separate male and female sides.

- 400 protrusions in a 20 x 20 pattern
- Black, polyolefin-based construction
- High-force attachment for interior components

Four layers define Twin Lock construction



Confirm the official product drawing and material declaration before customer release.

Item	Source-reported specification
Layer A	Polyolefin
Layer B	Acrylic foam
Layer C	Acrylic resin adhesive
Layer D	Release liner
Product thickness	3.6 +/- 0.2 mm
Engaged thickness	5.6 +/- 0.3 mm
Tensile strength	20-28 kgf on 25 x 25 mm

Engagement force changes with temperature

Configuration	23 C	-30 C	120 C
#400 / #400	26.3 kgf	33.7 kgf	8.7 kgf
#400 / #250	19.7 kgf	26.5 kgf	7.5 kgf

Average force

33.7

kgf at -30 C
#400 / #400

*Averages calculated
from four source
readings per condition.*

Average of four source readings per condition. Specimen: 25 x 25 mm.

Test method and rate should be confirmed from the original report.

Source reports heat and cycle-test passes



Heat test chamber



Environmental chamber



Loaded specimen test



Cycle-test setup

Source conditions

100 C oven test for 8 h | 3-cycle: 120 C / -40 C / 23 C at 50% RH, 8 h each

Applications extend beyond seating



Sunroof frame



Headliner / sunroof



Quarter glass



Door trim

Application-specific validation must cover the final substrate, primer, thermal environment and service load.

Production assets cover forming and finishing



Thermoforming



Slitting



Extrusion / coating



Web handling



Conversion



Multi-station line



Forming equipment



Production cells

North American RFQs need a qualification pack

SOURCE-REPORTED EVIDENCE

- ISO 9001:2015 status
- Hook separation-force results
- Hyundai-Kia MS test summary
- Twin Lock temperature and cycle data

CONFIRM BEFORE RFQ RELEASE

- IATF 16949, PPAP and IMDS capability
- FMVSS 302 and customer-specific tests
- Foam leakage, magnet seating and aging methods
- Annual capacity, lead time and U.S. landed price

Start with a controlled sampling program



Recommended first engagement

Define one application, one substrate stack and one customer test plan.

- Technical data review
- Production-intent samples
- Bench and foam-molding trial
- Commercial RFQ and approval plan

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